MA1513 Exam (Answers)

AY2024/2025 Semester I

MCQ/FIB (Q1-15; 3 marks each) [Correct answers are highlighted. Explanations in red.]

1. Given **Ax** = **b** is a non-homogeneous linear system where **A** is a 3 x 5 matrix. Suppose the general solution is given by $\begin{cases} x_1 &= s+2t \\ x_2 &= t-s-1 \\ x_3 &= s \\ x_4 &= t+2 \\ x_5 &= t \end{cases}$.

   Which of the following statements are correct? (Pick all correct options)

   i. The RREF of the augmented matrix (**A** | **b**) has exactly 2 non-pivot columns.
   ii. Any REF of the matrix **A** does not have a zero row.
   iii. $\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$ is a solution of **Ax** = **0**.
   iv. $\begin{pmatrix} 0 \\ 1 \\ 0 \\ -2 \\ 0 \end{pmatrix}$ is a solution of **Ax** = **b**.
   v. For any 3 x 1 column matrix **c**, the linear system **Ax** = **c** is consistent.

(i) is not correct. There are three non-pivot columns in the RREF of (A | b), including the augmented column.
(ii) In the REF of A, there are three pivot columns. So all the three rows of the REF must be non-zero rows.
(iii) The general solution of Ax = 0 is $\begin{cases} x_1 &= s+2t \\ x_2 &= t-s \\ x_3 &= s \\ x_4 &= t \\ x_5 &= t \end{cases}$. By letting s=0 and t = 1, we get the particular solution in (iii).
(iv) While $\begin{pmatrix} 0 \\ -1 \\ 0 \\ 2 \\ 0 \end{pmatrix}$ is a solution of Ax = b, the negative $\begin{pmatrix} 0 \\ 1 \\ 0 \\ -2 \\ 0 \end{pmatrix}$ is not a solution, as solutions for a non-homogeneous system are not closed under scalar multiplication.
(v) From (ii), we have REF of A does not have a zero row. This means REF of (A | c) does not have a row which correspond to all zero coefficients with non-zero constant term. Hence, Ax = c must be consistent.

2. Suppose **A** is a non-singular n x n matrix. Which of the following statements are true? (Pick all correct options)
   i. Any REF of $\mathbf{A}^T$ is non-singular.
   ii. **A** has n linearly independent eigenvectors.
   iii. The row space of **A** is $\mathbf{R}^n$.
   iv. The rank of $\mathbf{A}^k$ is n for any integer k > 1.
   v. We can express any column of **A** as a linear combination of the remaining columns of **A**.

(i) $\det(A^T) = \det(A)$ which is non-zero. If $\det(A^T) \neq 0$, then $\det(\text{REF of } A^T) \neq 0$. So REF of A^T is non-singular.
(ii) A has n linearly independent columns, but need not have n linearly independent eigenvectors, if it is not diagonalizable.
(iii) The RREF of A is the identity matrix, and its n rows is the standard basis for R^n. Hence the row space is equal to R^n.
(iv) Any power of A is an nxn non-singular matrix, and hence must be full rank (i.e. rank(A) = n).
(v) On the contrary, since all n columns of A are linearly independent, none of the columns can be expressed as a linear combination of the other columns of A.

3. Suppose **A** is a 2x2 matrix with determinant 2. Then
   i. $\det(-\mathbf{A}) = a$
   ii. $\det(2\mathbf{A}^T\mathbf{A}) = b$
   iii. $\det(4\mathbf{A}^{-1}) = c$

   where a = 2, b = 16, c = 8. (Fill in the blanks)

(i) $\det(-A) = (-1)^2\det(A) = 2$.
(ii) $\det(2A^TA) = 2^2\det(A^T)\det(A) = 4x2x2 = 16$
(iii) $\det(4A^{-1}) = 4^2\det(A)^{-1} = 16x2^{-1} = 8$

4. Which of the following vectors are coplanar with (1, 2, 3) and (3, 2, 1)? (Pick all correct answers.)
   i. (1, 3, 2)
   ii. Any linear combination of (1, 2, 3) and (2, 1, 3)
   iii. (2, 1, 0)
   iv. (2, 0, 2)
   v. Any linear combination of (1, 0, -1) and (0, 1, 2)

Vectors coplanar to (1,2,3) and (3,2,1) must be linear combinations of these two vectors. The vectors (1, 3, 2) in (i), (2, 1, 3) in (ii) and (2, 0, 2) in (iv) are not, while (2, 1, 0) in (iii) and (1, 0, -1), (0, 1, 2) in (v) are linear combinations of (1, 2, 3) and (3, 2, 1).

5. Which of the following is a basis for the subspace given by span{ (1, 0, 1, 1), (1, 2, 0, 1), (1, 4, -1, 1), (1, -2, 2, 1), (2, 2, 1, 1) }? (Pick all correct answers)
   i. { (1, 0, 1, 1), (1, 2, 0, 1), (1, 4, -1, 1) }
   ii. { (1, 4, -1, 1), (1, -2, 2, 1), (2, 2, 1, 1) }
   iii. { (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0) }
   iv. { (1, 2, 0, 1), (1, 4, -1, 1), (1, -2, 2, 1) }
   v. { (1, 0, 1, 1), (1, 4, -1, 1), (2, 2, 1, 1)}

Form the matrix $A = \begin{pmatrix} 1 & 1 & 1 & 1 & 2 \\ 0 & 2 & 4 & -2 & 2 \\ 1 & 0 & -1 & 2 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$ and reduce to RREF $R = \begin{pmatrix} 1 & 0 & -1 & 2 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

(i) Not a basis - The first three columns of R are not linearly independent.
(ii) A basis - The last three columns of R are linearly independent.
(iii) Not a basis – These three vectors do not span the column space of A. (None of the columns of A is a linear combination of these three vectors.)
(iv) Not a basis - The 2^{nd}, 3^{rd} and 4^{th} columns of R are not linearly independent.
(v) A basis - The 1^{st}, 3^{rd}, 5^{th} columns of R are linearly independent.

6. The coordinate vector of a vector **v** with respect to the basis S = { (1, 0, 1), (0, 1, 1), (1, 1, 0)} is given by $(\mathbf{v})_S = (3, 6, 9)$. Then the coordinate vector of **v** with respect to the basis T = { (2, 1, 0), (0, 2, 1), (1, 0, 2)} is given by $(\mathbf{v})_T = (a, b, c)$ where a = _5_, b = _5_, c = _2_ (FIB)

v = 3(1, 0, 1) + 6(0, 1, 1) +9(1, 1, 0) = (12, 15, 9)
Solve for (12, 15, 9) = a(2, 1, 0) + b(0, 2, 1) +c(1, 0, 2) gives a = 5, b = 5, c = 2.

7. Which of the following sets are linearly independent? (Pick all correct options)
   i. { (10, 20, 30, 40), (11, 22, 33, 44) }
   ii. { (1, 5, 1, 3), (2, 0, 2, 4), (0, 0, 0, 0) }
   iii. { (1, 1, 0, 0), (0, 0, 1, 1), (1, 0, 1, 0), (0, 1, 0, 1) }
   iv. { (1, 2, 3, 0), (2, 3, 1, 0), (3, 1, 2, 0) }
   v. { (1, 1, 1, 1), (2, 2, 0, 2), (3, 0, 3, 0), (0, 0, 0, 4) }

(i) Linearly dependent – both vectors are scalar multiples of (1, 2, 3, 4)
(ii) Linearly dependent – any set containing the zero vector is linearly dependent.
(iii) Linearly dependent – the sum of the first two vectors is equal to the sum of the last two vectors.
(iv) Linearly independent – Use the standard approach to set up a vector equation using the three vectors, and show that the system has only the trivial solution.
(v) Linearly independent – Stack the 4 vectors into a 4x4 matrix and check that the determinant is not zero.

8. Three of the following subspaces of $\mathbf{R}^3$ are the same as each other. Select the two subspaces that are not the same as the three.

i. Span{ (1, -1, 0), (1, 0, -1) }

ii. Nullspace of $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}$

iii. Row space of $\begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}$

iv. Column space of $\begin{pmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & -1 & 0 \end{pmatrix}$

v. Solution space of $x + 2y + 3z = 0$

$i \subseteq ii$

$iii \subseteq ii$

$(2, -1, 0)$

It will be easier to work with the matrix A = $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}$ in (ii). Note that rank(A) = 1, so nullspace of A has dimension 2.

(i) Multiplying the two spanning vectors $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ by A give the zero vector. So these two vectors belong to the nullspace of A. Since they are linearly independent, Span{ (1, -1, 0), (1, 0, -1) } has dimension 2, which is the same as the nullspace of A. So the two spaces are the same.

(iii) Multiplying the third row $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ (in column form) of the matrix by A does not give the zero vector. So the row space of this matrix is not the same as nullspace of A.

(iv) Multiplying the three columns $\begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ by A give the zero vector. So these three vectors belong to the nullspace of A. The dimension of the column space is 2. So the column space of this matrix is the same as the nullspace of A.

(v) The vectors in (i) does not satisfy the equation in (v). So the solution space is not the same as (i), (ii) and (iv).

9. Let T: $\mathbf{R}^3 \to \mathbf{R}^3$ be a linear transformation such that $T\left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $T\left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

Then $T\left(\begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}\right) = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where a = _3_, b = _5_, c = _2_ (FIB)

$$\begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + 3\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \to T\left(\begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}\right) = 2\,T\left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right) + 3\,T\left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right) = 2\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 3\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 2 \end{pmatrix}.$$

10. Given **A** is a 3 x 3 diagonalizable matrix with eigenvalues 1 and 3 with corresponding eigenspaces E_1 = span$\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}\right\}$ and E_3 = span$\left\{\begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$

Which of the following matrices diagonalize **A**? (Pick all correct answers)

i. $\begin{pmatrix}0 & 1 & 1\\0 & 1 & 0\\1 & 1 & 0\end{pmatrix}$ — 1^{st} column is in E_3. 2^{nd} column is in E_1. 3^{rd} column is a linear combination of the two vectors in E_3.

ii. $\begin{pmatrix}2 & 4 & 0\\0 & 4 & 0\\2 & 4 & 6\end{pmatrix}$ — 1^{st} and 3^{rd} columns are scalar multiples of vectors in E_3. 2^{nd} column is scalar multiple of vector in E_1.

iii. $\begin{pmatrix}1 & 1 & 2\\1 & 0 & 2\\1 & 1 & 2\end{pmatrix}$ — 1^{st} and 3^{rd} columns are scalar multiple of each other, and hence not linearly independent.

iv. $\begin{pmatrix}2 & 1 & 0\\1 & 0 & 0\\2 & 1 & 1\end{pmatrix}$ — 1^{st} column is a linear combination of a vector in E_1 and a vector in E_3, which is not an eigenvector.

v. $\begin{pmatrix}1 & 0 & 0\\0 & 0 & 0\\1 & 1 & 0\end{pmatrix}$ — 3^{rd} column is a zero vector, which is not an eigenvector.

11. Given a 4x4 matrix **A** with characteristic polynomial $x(x-1)^a(x+1)^b$ for some positive integers a and b. Which of the following statements are true? (Pick all correct answers)

$r = 0 \quad V = 0$

$= \text{nullsp } A$

i. The nullity of **A** is 1.
ii. The dimension of one of the eigenspaces of **A** is 2.
iii. Exactly one of the eigenvalues of A has multiplicity 2.
iv. **A** cannot be diagonalizable.
v. det(**A**) = 0.

(i) Nullspace of A is the same as the eigenspace of A with eigenvalue 0. Since the multiplicity of eigenvalue 0 is 1 (from the characteristic polynomial), the dimension of the eigenspace is 1. So the nullity is 1.
(ii) It is possible that dimensions of all the eigenspaces are 1, regardless of the multiplicities.
(iii) Since the multiplicities must add up to 4, one of a and b must be 2.
(iv) If the dimension of the eigenspace is equal to the multiplicity of the corresponding eigenvalue (for both 1 and -1), then A will be diagonalizable.
(v) Since 0 is an eigenvalue of A, it is singular and det(A) = 0.

12. Given a 3 x 3 SDE $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}$ with fundamental set of solutions $\boldsymbol{x}_1 = e^t\begin{pmatrix}0\\1\\-1\end{pmatrix}, \boldsymbol{x}_2 = e^{-t}\begin{pmatrix}0\\1\\1\end{pmatrix}, \boldsymbol{x}_3 = \begin{pmatrix}1\\1\\1\end{pmatrix}$. Supposed $y(0) = \begin{pmatrix}3\\6\\4\end{pmatrix}$. Then the particular solution of the initial value problem is
$\boldsymbol{y} = a\boldsymbol{x}_1 + b\boldsymbol{x}_2 + c\boldsymbol{x}_3$ where a = _1_, b = _2_, c = _3_ (FIB)

Solving $\begin{pmatrix}3\\6\\4\end{pmatrix} = a\begin{pmatrix}0\\1\\-1\end{pmatrix} + b\begin{pmatrix}0\\1\\1\end{pmatrix} + c\begin{pmatrix}1\\1\\1\end{pmatrix}$, gives a = 1, b = 2 and c = 3.

13. An SDE is given by $\begin{pmatrix} {y_1}' \\ {y_2}' \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & -4 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$. Which of the following are solutions of the SDE?

i. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t}\begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}$

ii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t}\begin{pmatrix} -1/2 \\ 0 \end{pmatrix}$

iii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t}\begin{pmatrix} -t-1 \\ t \end{pmatrix}$

iv. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t}\begin{pmatrix} -t \\ t-\frac{1}{2} \end{pmatrix}$

v. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{-2t}\begin{pmatrix} t-\frac{1}{2} \\ -t \end{pmatrix}$

$\begin{pmatrix} 0 & 2 \\ -2 & -4 \end{pmatrix}$ has one eigenvalue -2 with one linearly independent eigenvector $\boldsymbol{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

We find the generalised eigenvector for $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ by solving $\begin{pmatrix} 2 & 2 \\ -2 & -2 \end{pmatrix}\boldsymbol{u} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ which gives solutions $\boldsymbol{u} = \begin{pmatrix} -\frac{1}{2} - s \\ s \end{pmatrix}$ for any s.

(i) $\frac{1}{2}\boldsymbol{v} = \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}$ is an eigenvector, so $\frac{1}{2}\boldsymbol{v}e^{-2t}$ is a solution of the SDE.

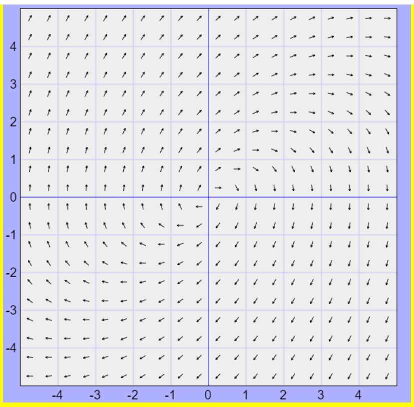
(ii) $\boldsymbol{u} = \begin{pmatrix} -1/2 \\ 0 \end{pmatrix}$ is a generalised eigenvector (with s = 0) of $\boldsymbol{v}$, but $\boldsymbol{u}e^{-2t}$ is not a solution of the SDE.

(iii) $\boldsymbol{w} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ is not a generalised eigenvector of $\boldsymbol{v}$, so $(t\boldsymbol{v} + \boldsymbol{w})e^{-2t}$ is not a solution of the SDE.

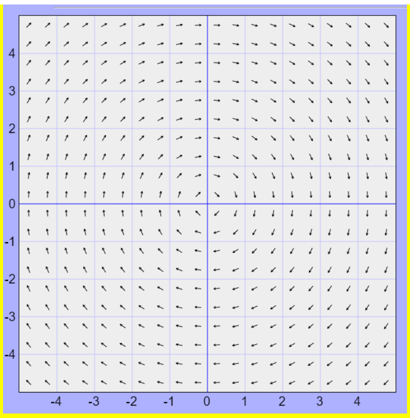
(iv) $\boldsymbol{u} = \begin{pmatrix} 0 \\ -1/2 \end{pmatrix}$ is a generalised eigenvector (with s = $-1/2$) of $\boldsymbol{v}$, so $(t\boldsymbol{v} + \boldsymbol{u})e^{-2t}$ is a solution of the SDE.

(v) $\boldsymbol{u} = \begin{pmatrix} -1/2 \\ 0 \end{pmatrix}$ is a generalised eigenvector (with s = 0) of $\boldsymbol{v}$ but not - $\boldsymbol{v}$, so $(-t\boldsymbol{v} + \boldsymbol{u})e^{-2t}$ is not a solution of the SDE.

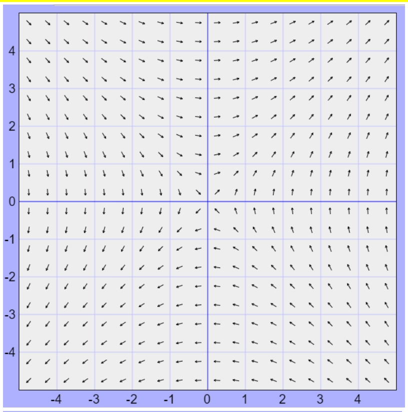
14. Which of the following are the possible direction fields for an SDE **y**′ = $\begin{pmatrix} 0 & a \\ -1 & b \end{pmatrix}$**y** for some a and b ?

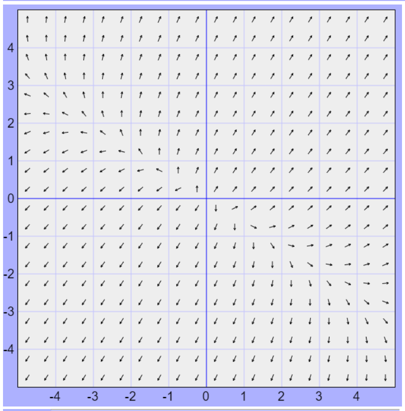
i. 

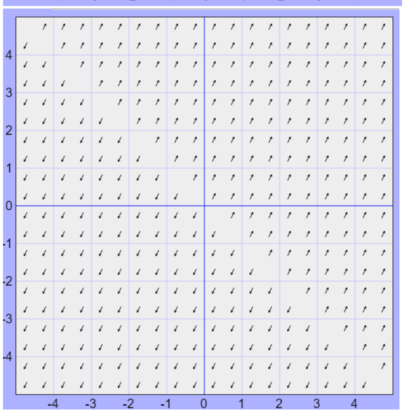
e.g. $\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$

ii. 

e.g. $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

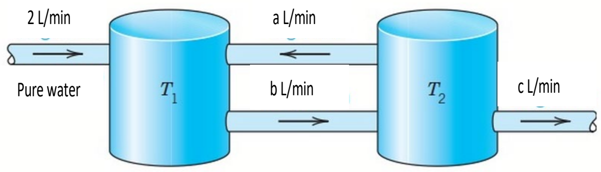
iii. 

iv. 

v. 

Multiplying $\begin{pmatrix} 0 & a \\ -1 & b \end{pmatrix}$ to the test point $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ gives the tangent vector $\begin{pmatrix} 0 \\ -1 \end{pmatrix}$, which is an arrow that is pointing vertically downward. Looking at the direction field near the point (1, 0), only (i) and (ii) gives matching tangent vectors.

15. Consider the two tanks in the diagram. Tank 1 holds 100 litre of salt water and tank 2 holds 200 litre of salt water.



Suppose the amount of salt in tank 1 and 2 are $y_1(t)$ and $y_2(t)$ respectively and the SDE that describe the amount of salt in the two tanks is given by $\begin{pmatrix} y_1{}' \\ y_2{}' \end{pmatrix} = \frac{1}{100}\begin{pmatrix} -6 & 2 \\ 6 & -3 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$.

Then a = _4_, b = _6_, c = _2_ (FIB).

Tank 1: Rate of inflow is $\frac{a}{200}y_2$ and rate of outflow is $\frac{b}{100}y_1$.
Tank 2: Rate of inflow is $\frac{b}{100}y_1$ and rate of outflow is $\frac{a+c}{200}y_2$.
Comparing with the underlying matrix $\frac{1}{100}\begin{pmatrix} -6 & 2 \\ 6 & -3 \end{pmatrix}$, we have b = 6, a = 4 and a + c = 6, which gives c = 2.

Written Questions (Q16 – Q19)

Q16. [10 marks]

A manufacturer makes four types of furniture A, B, C, D. The company has available 2600 units of wood, 600 units of upholstery and 2400 units of labour. The manufacturer wants a production schedule that uses all of these resources. The various products require the following amounts of resources.

	A	B	C	D
Wood	4	4	3	4
Upholstery	0	1	0	2
Labour	2	3	4	4

(i) Set up a linear system with variables a, b, c, d, which represent the number of pieces of products A, B, C, D respectively to be manufactured.

Ans: $\begin{cases} 4a + 4b + 3c + 4d = 2600 \\ b + 2d = 600 \\ 2a + 3b + 4c + 4d = 2400 \end{cases}$

(ii) Solve the linear system in (i).

Ans: RREF = $\begin{pmatrix} 1 & 0 & 0 & -1 & -100 \\ 0 & 1 & 0 & 2 & 600 \\ 0 & 0 & 1 & 0 & 200 \end{pmatrix}$. General solution: $\begin{cases} a = -100 + t \\ b = 600 - 2t \\ c = 200 \\ d = t \end{cases}$.

(iii) What is the range for the possible values of d?

Ans: Since all the variables should have non-negative values,

$$\begin{aligned} a &= -100 + t \geq 0 \to t \geq 100 \\ b &= 600 - 2t \geq 300 \geq t \\ c &= 200 \\ d &= t \geq 0 \end{aligned}$$

So the range of t is given by $100 \leq t \leq 300$.

Q17. [10 marks]

Let A = $\begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & -2 \\ 1 & 2 & 2 & 1 \end{pmatrix}$ and b = $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$.

(i) Find a basis for and the dimension of the column space of A.

Ans: RREF(A) = $\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. So a basis is $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ and the dimension = 2.

(ii) Find the projection of b onto the column space of A.

Ans: Least squares solution of Ax = b is $\begin{cases} x & = & \frac{2}{3} + t - 2s \\ y & = & s \\ z & = & -t \\ w & = & t \end{cases}$ where s, t are parameters. Take s and t to be 0, we get x_0 = $\begin{pmatrix} 2/3 \\ 0 \\ 0 \\ 0 \end{pmatrix}$. The projection p = Ax_0 = $\begin{pmatrix} 2/3 \\ 4/3 \\ 2/3 \end{pmatrix}$.

(iii) Find a unit vector in R^3 which is orthogonal to all the vectors in the column space of A.

Ans: p – b = $\begin{pmatrix} -4/3 \\ 1/3 \\ 2/3 \end{pmatrix}$. So a unit vector is $\begin{pmatrix} -4/\sqrt{21} \\ 1/\sqrt{21} \\ 2/\sqrt{21} \end{pmatrix}$ or $\pm\begin{pmatrix} -0.8729 \\ 0.2182 \\ 0.4364 \end{pmatrix}$.

Q18. [12 marks]

A certain species of animals has two life stages: juvenile (up to 1 year old) and adult. Each year,

- an adult produces an average of 1.2 juveniles;
- 50% of the juveniles survive to become adults;
- 40% of the adults survive.

Let $x_n = \begin{pmatrix} j_n \\ a_n \end{pmatrix}$ where j_n and a_n are the number of juveniles and adults respectively in year n.

(i) Find the stage matrix A such that $x_{n+1} = Ax_n$.

Ans: The three points above translate into a system of two equations:

$$\begin{array}{rcrcl} j_{n+1} & = & & & 1.2a_n \\ a_{n+1} & = & 0.5j_n & + & 0.4a_n \end{array} \text{ or } \begin{pmatrix} j_{n+1} \\ a_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 1.2 \\ 0.5 & 0.4 \end{pmatrix} \begin{pmatrix} j_n \\ a_n \end{pmatrix}.$$

So $A = \begin{pmatrix} 0 & 1.2 \\ 0.5 & 0.4 \end{pmatrix}$.

(ii) Diagonalize the matrix A^n.

Ans: Find the eigenvalues -0.6 and 1, and corresponding eigenvectors $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 6/5 \\ 1 \end{pmatrix}$.

(Note that students are allowed to use MATLAB to find the eigenvalues and eigenvectors without showing working. The above eigenvectors can be replaced with any scalar multiples.)

Form the matrix P using the eigenvectors: $P = \begin{pmatrix} -2 & 6/5 \\ 1 & 1 \end{pmatrix}$.

For the matrix D using the eigenvalues: $D = \begin{pmatrix} -0.6 & 0 \\ 0 & 1 \end{pmatrix}$.

(Note that the order of the two columns of P and D can be swopped correspondingly.)

Then $A^n = \begin{pmatrix} -2 & 6/5 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} (-0.6)^n & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 6/5 \\ 1 & 1 \end{pmatrix}^{-1}$.

(iii) Suppose the initial populations j_0 and a_0 for juveniles and adults are respectively 10,000 and 8,000. Will the populations be stabilized in the long term? Why?

Ans: Yes. In the long run $A^n \to \begin{pmatrix} -2 & 6/5 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 6/5 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 3/8 & 3/4 \\ 5/16 & 5/8 \end{pmatrix}$

and the population stablilizes to $\begin{pmatrix} 3/8 & 3/4 \\ 5/16 & 5/8 \end{pmatrix} \begin{pmatrix} 10000 \\ 8000 \end{pmatrix} = \begin{pmatrix} 9750 \\ 8125 \end{pmatrix}$.

(iv) True or false: Regardless of the initial populations j_0 and a_0, the ratio of juvenile to adult populations will stabilize in the long term. Justify your answer.

Ans: True.

Long term population: $\begin{pmatrix} 3/8 & 3/4 \\ 5/16 & 5/8 \end{pmatrix} \begin{pmatrix} j_0 \\ a_0 \end{pmatrix} = \begin{pmatrix} 9/8(j_0 + a_0) \\ 15/16(j_0 + a_0) \end{pmatrix}$.

So the ratio is always 9/8 : 15/16, or 6 : 5 regardless of the initial populations j_0 and a_0.

Q19. [13 marks]

A particular solution of an SDE $y' = Ay$ is given by: $y = 3e^{-t}\begin{pmatrix}-\cos t + \sin t\\ 2\cos t\end{pmatrix} - 3e^{-t}\begin{pmatrix}-\sin t - \cos t\\ 2\sin t\end{pmatrix}$.

(i) Find the matrix A.

Ans: Eigenvalues $r_1 = -1 + i$, $r_2 = -1 - i$. Eigenvectors $\mathbf{v}_1 = \begin{pmatrix}-1\\ 2\end{pmatrix} + i\begin{pmatrix}-1\\ 0\end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix}-1\\ 2\end{pmatrix} - i\begin{pmatrix}-1\\ 0\end{pmatrix}$

P = $\begin{pmatrix}-1-i & -1+i\\ 2 & 2\end{pmatrix}$; **D** = $\begin{pmatrix}-1+i & 0\\ 0 & -1-i\end{pmatrix}$.

A = PDP^{-1} = $\begin{pmatrix}0 & 1\\ -2 & -2\end{pmatrix}$

(ii) State the type and stability of the equilibrium point.

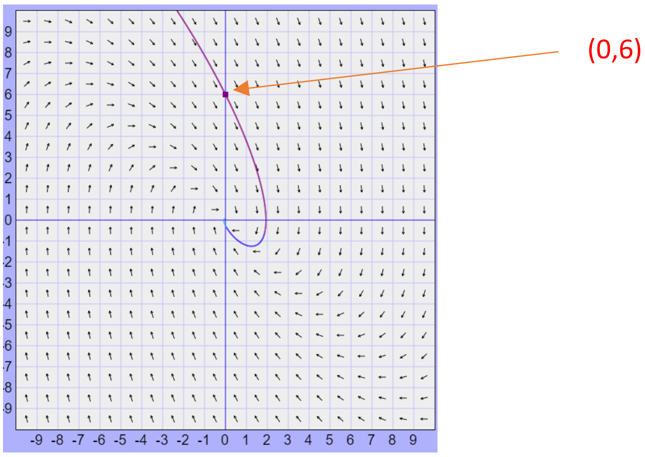
Ans: Spiral; asymptotically stable

(iii) Sketch the trajectory of the given particular solution on the phase plane, indicating on the plane the initial value when t = 0.

Ans:

When t = 0, $\begin{pmatrix}y_1\\ y_2\end{pmatrix} = 3\begin{pmatrix}-1+0\\ 2\end{pmatrix} - 3\begin{pmatrix}-0-1\\ 0\end{pmatrix} = \begin{pmatrix}0\\ 6\end{pmatrix}$.

Spiral clockwise inwards, passing through the point (0, 6).



(iv) What is the tangent vector of the trajectory at the point when it first crosses the horizontal y1-axis after t = 0?

Ans: When the trajectory crosses the y_1-axis, the coordinate $y_2 = 0$.

So we set $3e^{-t}\begin{pmatrix}-\cos t + \sin t\\ 2\cos t\end{pmatrix} - 3e^{-t}\begin{pmatrix}-\sin t - \cos t\\ 2\sin t\end{pmatrix} = \begin{pmatrix}y_1\\ 0\end{pmatrix}$ and solve for t using the y_2 component:

$$6e^{-t}(\cos t - \sin t) = 0 \Longrightarrow \cos t = \sin t \Longrightarrow \frac{\sin t}{\cos t} = 1 \Longrightarrow \tan t = 1\,.$$

So the trajectory first crosses the y_1-axis after $t = 0$ when $t = \pi/4$ (i.e. 45°), at the point

$$3e^{-\frac{\pi}{4}}\begin{pmatrix}-\cos\frac{\pi}{4}+\sin\frac{\pi}{4}\\ 2\cos\frac{\pi}{4}\end{pmatrix} - 3e^{-\frac{\pi}{4}}\begin{pmatrix}-\sin\frac{\pi}{4}-\cos\frac{\pi}{4}\\ 2\sin\frac{\pi}{4}\end{pmatrix}$$

$$= 3e^{-\pi/4}\left[\begin{pmatrix}0\\ \sqrt{2}\end{pmatrix}-\begin{pmatrix}-\sqrt{2}\\ \sqrt{2}\end{pmatrix}\right] = 3\sqrt{2}e^{-\pi/4}\begin{pmatrix}1\\0\end{pmatrix}.$$

And the tangent vector at this point is given by $3\sqrt{2}e^{-\pi/4}\begin{pmatrix}0 & 1\\ -2 & -2\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = 3\sqrt{2}e^{-\pi/4}\begin{pmatrix}0\\-2\end{pmatrix}.$